

AI 507: Artificial Intelligence and Society



#6 Human-Robot Interaction

Associate Prof. Dr. Lena Frischlich

Flashlight

Flashlight

Innovation

Economic perspectives

Social innovation

Big No's: Uneffective, harmful, unethical

Social innovation compass

Value to Society

Efficacy

Impact magnitude

Scalability

Mission alignment

Social/ environmental consequences

Failed innovations

AI Valgkompass

Dissemination of innovation

Diffusion of Innovations

Theory of planned behaviour

Attitudes ("will")

Perceived norms ("should")

Behavioural control ("can")

Technology acceptance model

Perceived usefulness

Perceived ease of use

Unified technology acceptance model

Trust

Cognitive similarity

Generative AI & Journalism

Stages of media work

News gathering

News composition

News distribution

AI + investigative journalism

AI + fact-checking

AI for news in Denmark & beyond

**Questions: Send me an
itslearning message**



Then let's get started!

After today's video lecture, you will hopefully...

- ☐ ...have initial insights into human's perceptions of artificial agents
- ☐ ...are familiar with the uncanny valley effect
- ☐ ...have learned more about human-robot relationships

Robots are based on automation

Automation

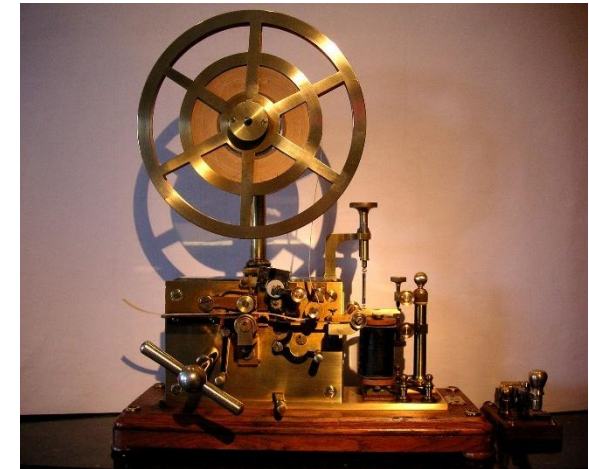
= Transfer of functions of the production process from humans to artificial systems

Comprises:

- *Mechanization*, in which machines provide the energy for the production process, but humans act (e.g. "Chess Turk")
- *Machinization*, in which machines (partially) take over the process (e.g., telegraphy))
- Today: Machines take over control and regulation tasks (usually through certain feedback loops))
- Increasing level of autonomy

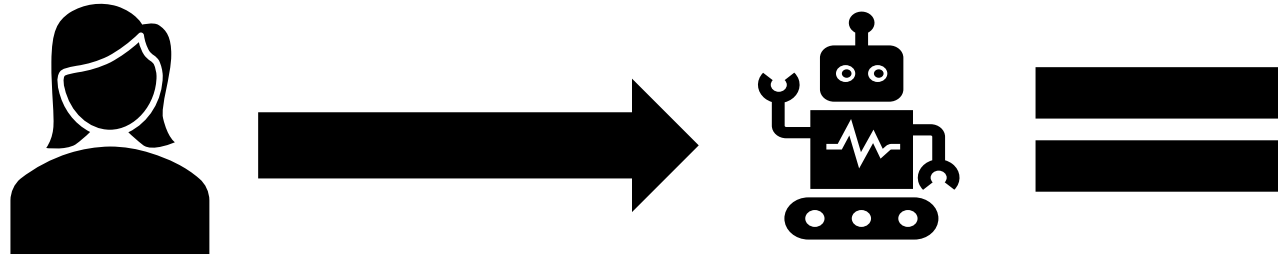


Kupferstich The Turk Racknitz (1789)
via Wikimedia

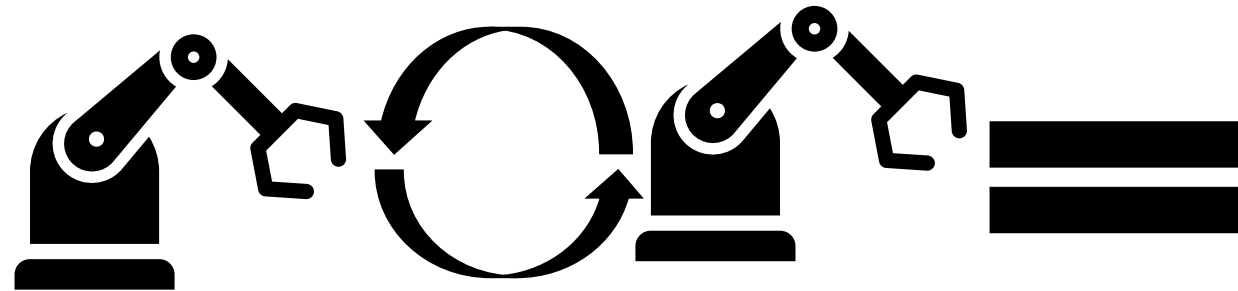


Ericsson Telegraf Serie 800,
ca.1880 Holger Ellgaard via
Wikimedia

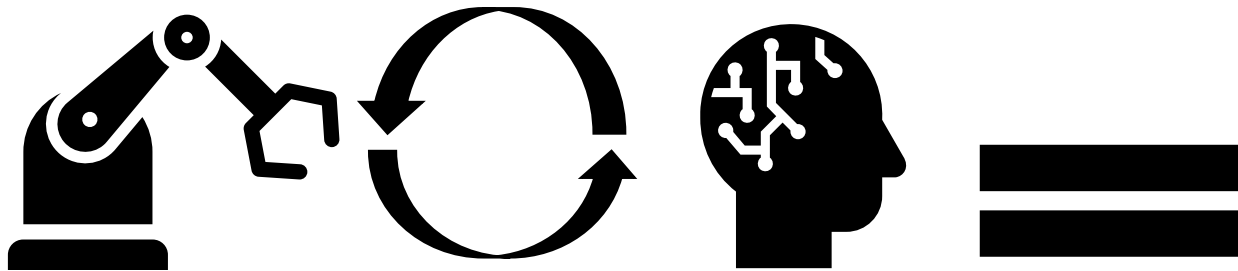
This is also reflected in central terms



Avatar



Agent



Embodied/
anthromorph agent

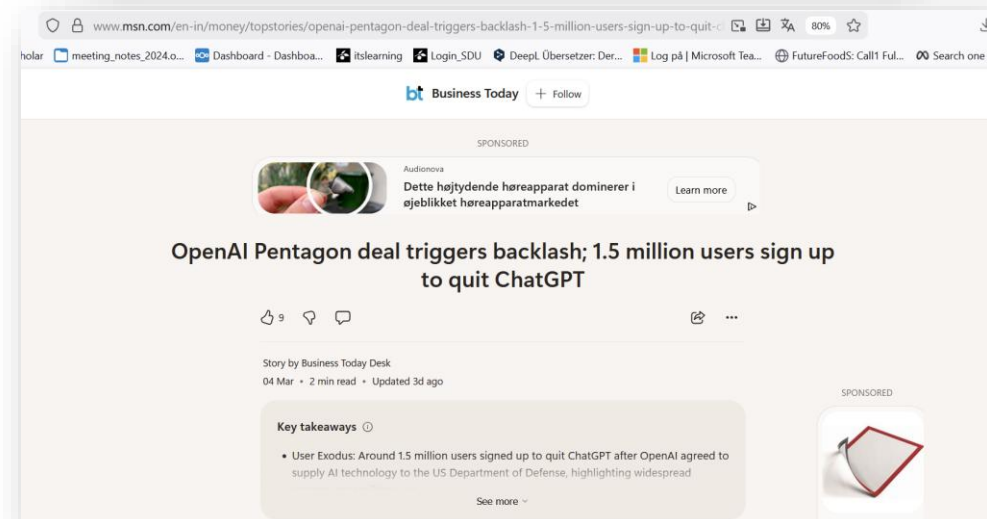
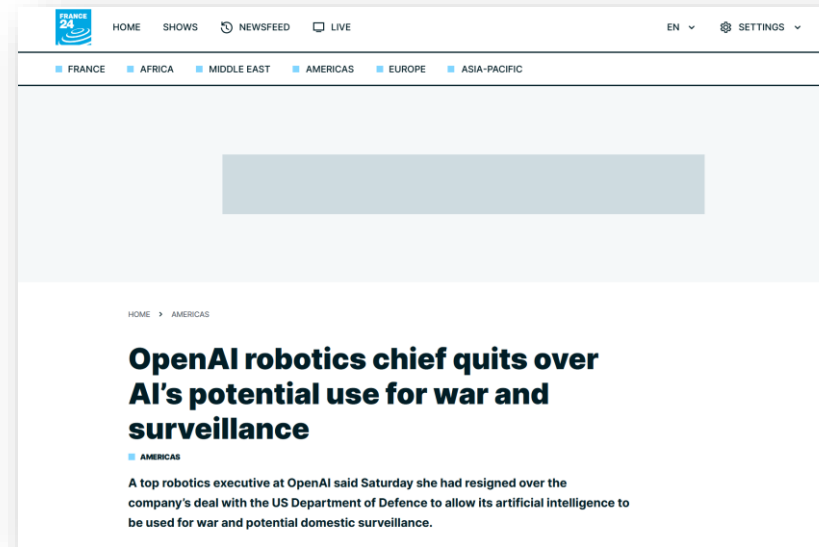
One central area of debate – autonomous weapon systems



<https://www.youtube.com/watch?v=VqCMkXftDx8>

Conflict about agentic warfare

- Since 2018, UN Secretary-General António Guterres argues that lethal autonomous weapons systems are unacceptable and morally repugnant
- UN calls for their prohibition under international law or at least for augmentation (avatar) rather than replacement (agent)
- At the heart of the Anthropic vs Department of War controversy



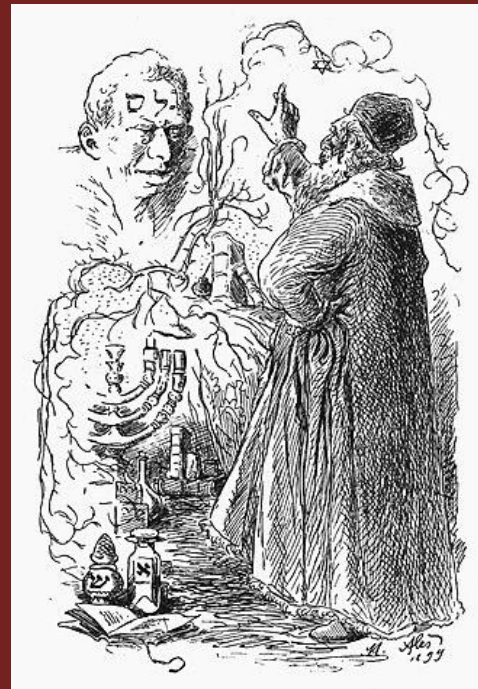
The dream of the artificial other is old

From Ancient Greek sculpturer
Pygmalion who falls in love with his
statue and brings her to life



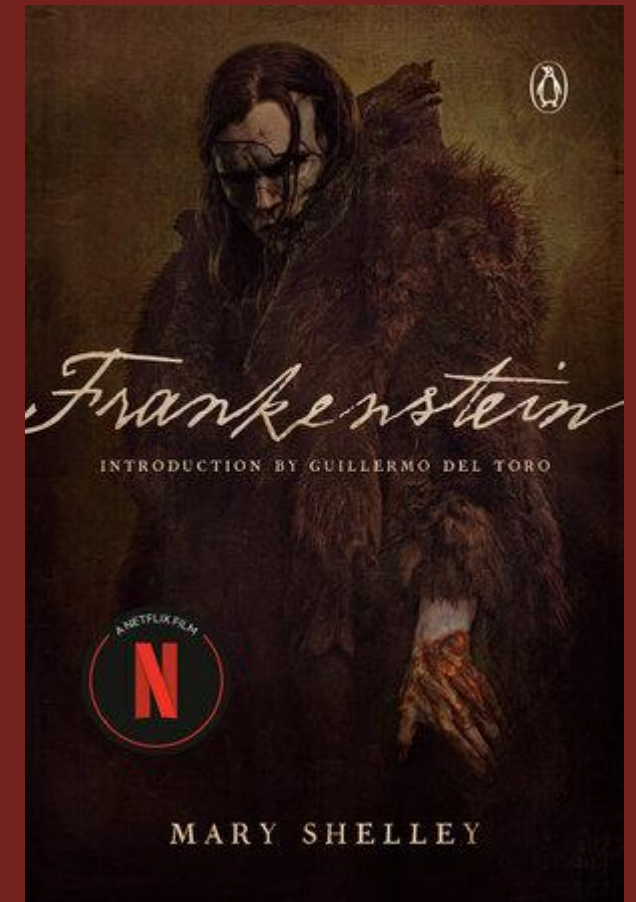
[https://de.wikipedia.org/wiki/Pygmalion#/media/Datei:Pygmalion_\(Baur\).jpeg](https://de.wikipedia.org/wiki/Pygmalion#/media/Datei:Pygmalion_(Baur).jpeg)

Over Jewish myths of the Golem
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<https://en.wikipedia.org/wiki/Golem>

To the horror of Frankenstein...

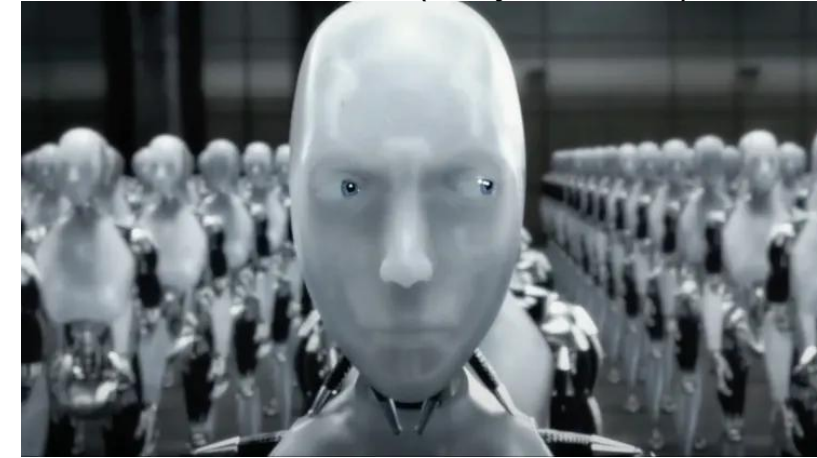


The fathers of robotics

(Asimov, 1994; KE-Next, 2023)

- In 1942, the Russian sci-fi writer Isaak Asimov created the iconic „three laws of robots“ inspiring stories til today
 - A robot must not injure a human being or cause harm through inaction
 - A robot must obey the commands of humans – as long as you are not in conflict with the first law
 - A robot must protect its own existence as long as this protection does not contradict the first or second law
- Joseph Engelberger read this story as a boy and started dreaming of his personal robots since then
- Together with George Devol, who had patented a machine control process in 1954, Engelberger founded the first robotics company (Unimation)

Scene from I, Robot (Proyas, 2004)

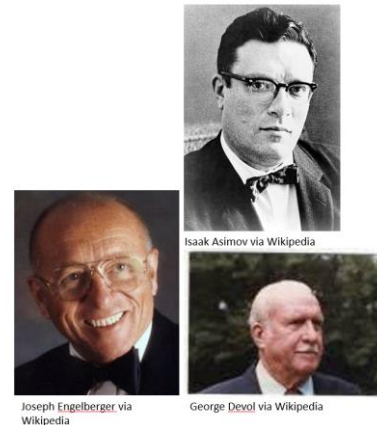


<https://www.fastcompany.com/91152591/i-robot-will-smith-ai-20-years-later>

I'm Your Man (Schrader, 2021)



https://www.rottentomatoes.com/m/im_your_man_2021



Joseph Engelberger via Wikipedia

Isaak Asimov via Wikipedia

George Devol via Wikipedia

And today?

(Richter, 2023; <https://investinodense.dk/robotics-automation/>)

3.5 million
Industrial robots
worldwide
-more than ever
before

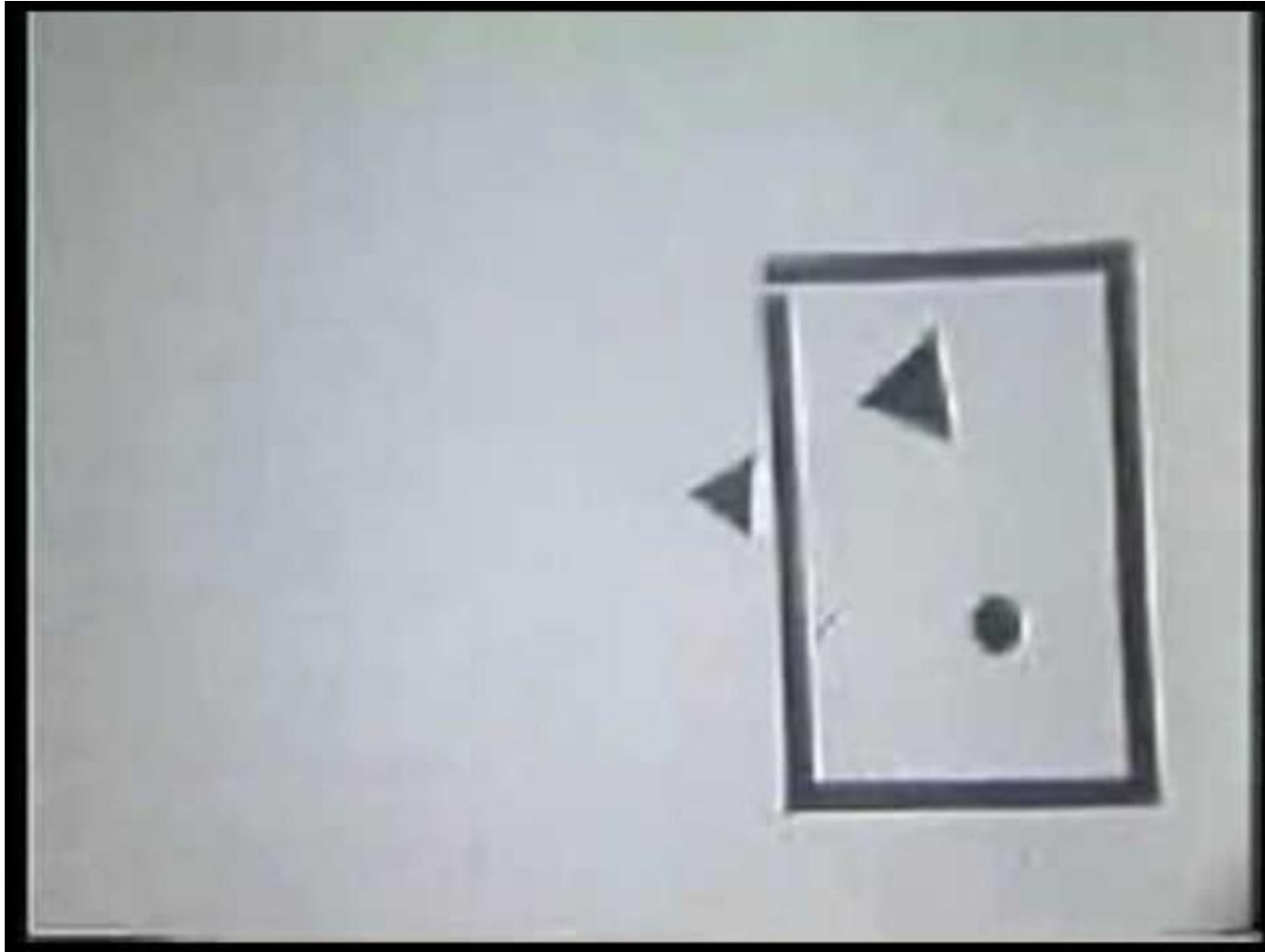
5 key trends (according to the International Federation of Robotics))

1. Increasingly easy to use.
2. Artificial Intelligence (Smartification)
3. Relocation of production to Europe/the USA
4. Energy efficiency (e.g. lower heating costs but also recovery of kinetic energy)
5. Sustainability and Prepare-to-Repair.

Odense is at the forefront of these developments with over 160 companies working in the industry

How do humans perceive artificial others?

What happens here?



Haider & Simmel, 1944, <https://www.youtube.com/watch?v=VTNmLt7QX8E>

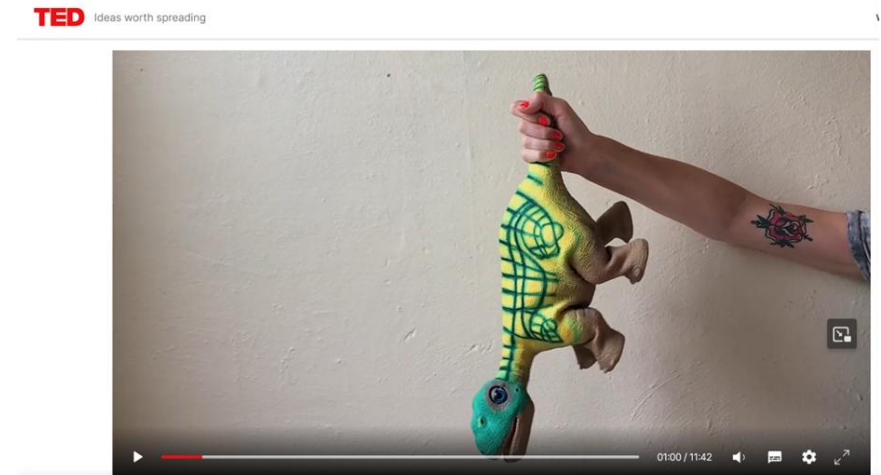
What happens
here?

- Remember social cognition? Social perception is our „default“ activity
- We *attribute* intention to independently moving objects
- This also extends to automated actors

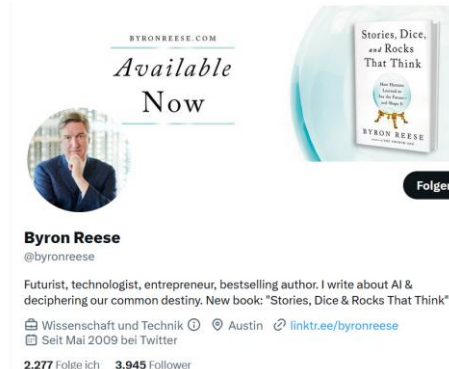
The media equation

(Reeves & Nass, 1996; Shechtman & Horowitz, 2003)

- Basic idea – Humans treat computers (and also robot dinosaurs) like other social actors.
- Tested in various so-called “CASA” experiments (Computers-as-Social-Actors paradigm).
- Basic logic: If people show the same social behaviours towards computers as towards humans, then they perceive computers as social actors
- Diverse experiments show, f. ex., that we are polite towards robots – but also that we feel obliged to respond to robots’ politeness



https://www.ted.com/talks/kate_darling_why_we_have_an_emotional_connection_to_robots



Clifford Nass via Wikipedia

Extensions of CASA

(Blascovich, 2002; Krämer & van der Pütten, 2015; Nass & Moon, 2000)

Ethopoeia Approach

- Minimal social cues are sufficient for social perceptions
- E.g. smiling back

Threshold model of social influence (TMSI)

- Two key factors.
 - Agency
 - Behavioural realism
- If agency is high (= agent): behavioural realism is not so important
- If agency is low (= avatar): behavioural realism is key
- Later studies showed that behavioural realism and social cues matter (more is better)

Let's look at one example study (Shi et al. 2015)

890

- Deciding how to approach someone and open the conversation is difficult for humans
- But it is a real challenge for robots.
 - Do I speak to a customer?
 - How close can I get
 - When do I start talking?
- Observations showed: Sales assistants approached customers when customers were looking in their direction and were close to a product (→ spatial position)
- At ~70 cm, the conversation was opened
- Overall: More positive evaluation than "general greetings" or "always stand by the product and explain"

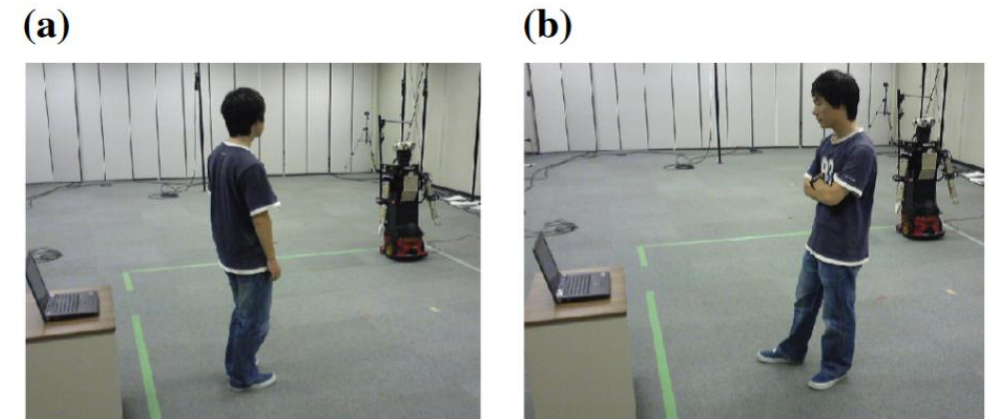


Fig. 1 Situations in a shop. **a** Looking at robot. **b** Looking at a product



https://www.tiktok.com/@kings_sushi/video/7278626000295005473

But there are limits: The uncanny valley effect

(Mori, 1970; Mara et al., 2022)

Masahiro Mori

M. Mara et al., Uncanny Valley Meta-Analysis

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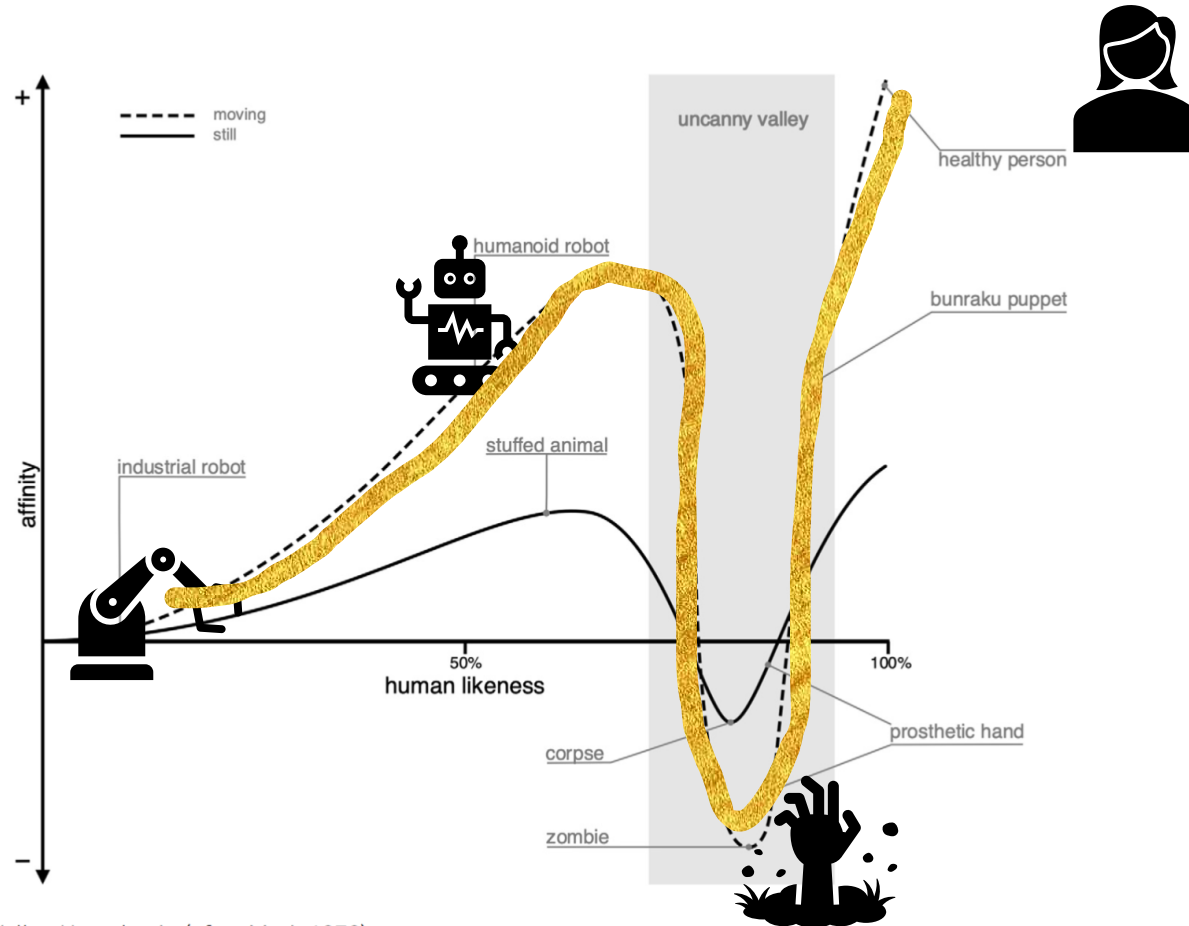


Figure 1. Uncanny Valley Hypothesis (after Mori, 1970).

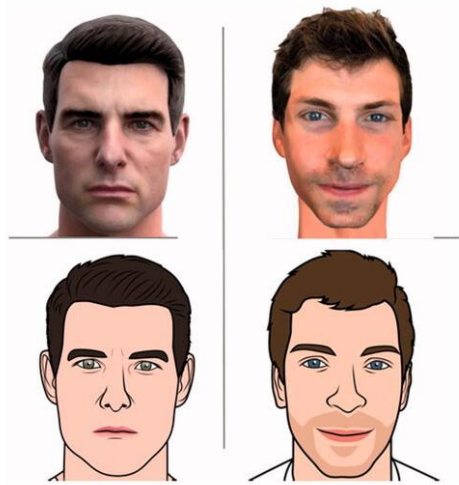


Mounting evidence

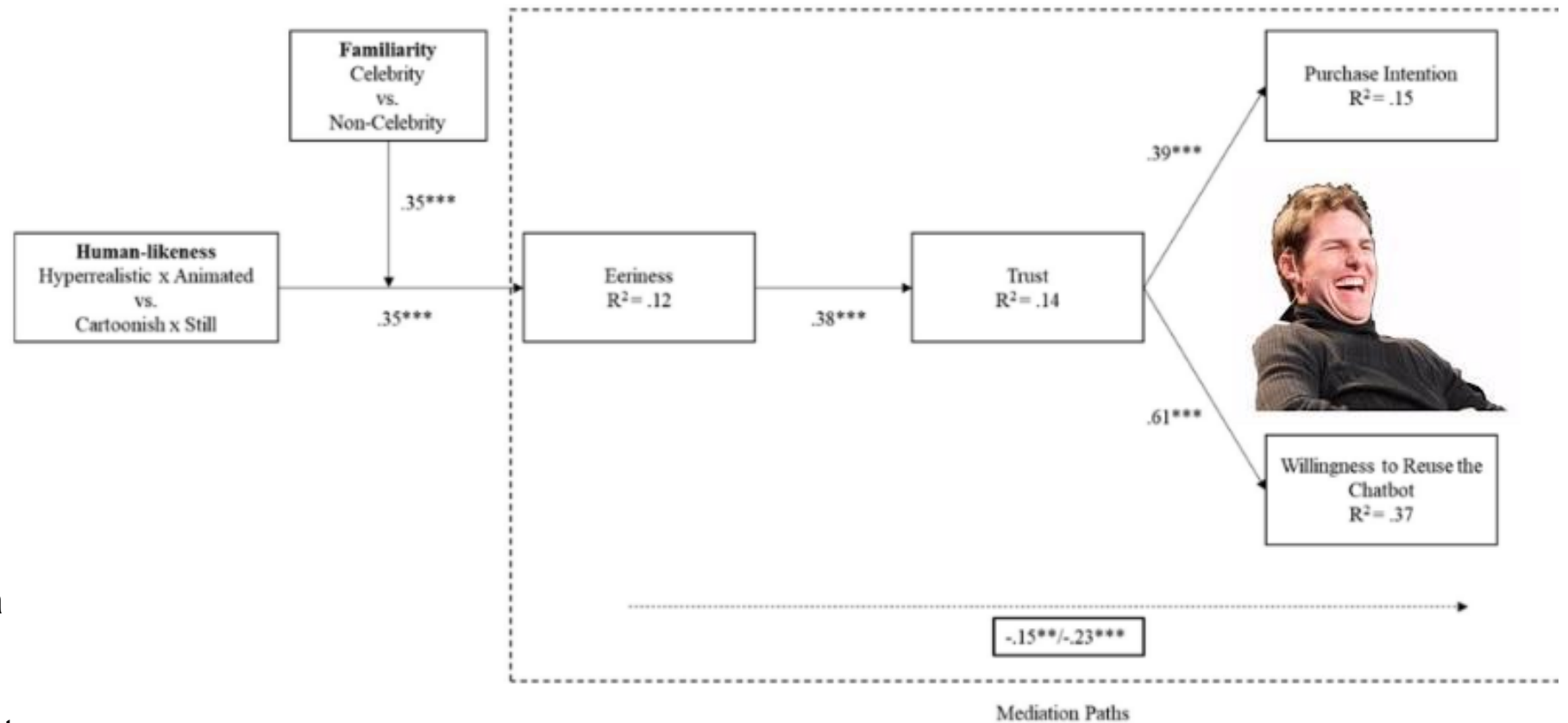
(Appel et al. 2020; Mara et al., 2022)

- 49 studies with 131 reviews of robots
- Most robots are rated as moderately human-like
- Most reviews are moderately positive
- Confirmed: Static robots are perceived as less human-like
- Communicating robots are rated more positively
- More similarity, better ratings
- BUT– no statements can (yet) be made about the very human-like range
- Possibly: Rather, the perceived "consciousness" of the robot (attribution of experiencing & thinking) that triggers the effect

How to overcome the uncanny valley effect



Display full size



- 187 students from the Netherlands
- Chatbot to support buying a laptop
- Dependent variables: eeriness, trust in the chatbot, intentions to use the bot
- Intentions to buy the laptop

How to overcome the uncanny valley effect?

(Grundke et al. 2023)

Imagine a new robot, Elix.

Elix is a tool with arms and hands and can grab and carry things

Elix supports people in everyday life and can execute user commands

Imagine a new robot, Elix.

Elix is a tool with arms and hands and can grab and carry things

Elix can feel hunger, pain and pleasure and has consciousness and personality

In the next step, the subjects watched a video of Ellix

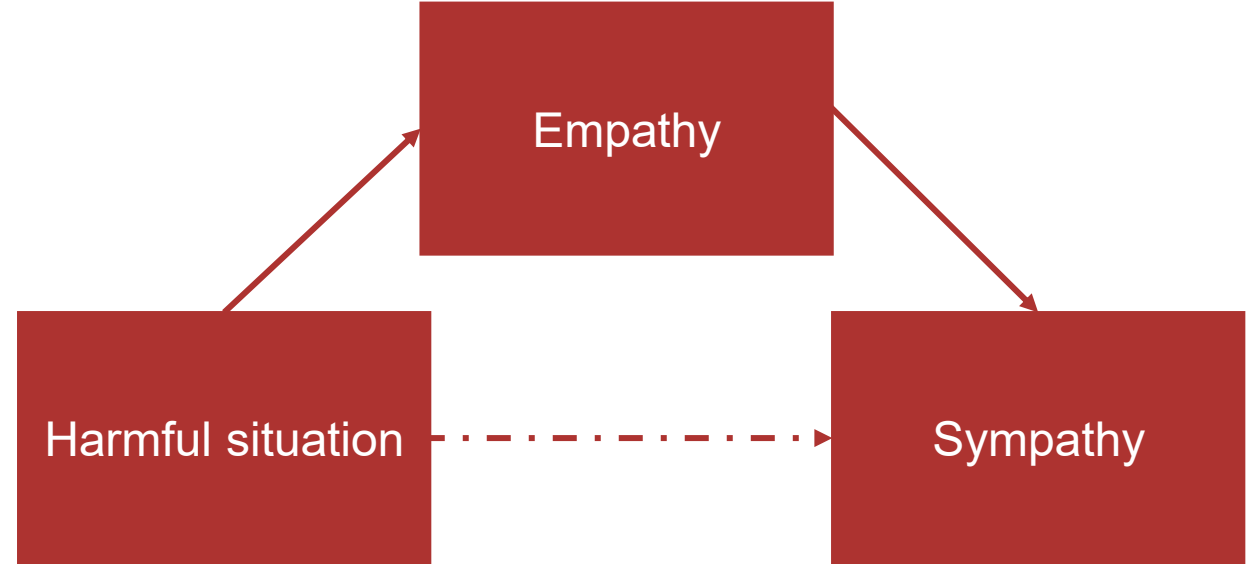
(Grundke et al. 2023)

The robot tries to lift a box and is stopped by an adult with a hockey stick, who constantly pushes the box to the side

At the end of the video, the adult Ellix falls to the ground with the hockey stick

Ellix remains lying on his back

What do you think?



However, not all emotions are perceived as equal

(Craig & Edwards, 2021)

- $N = 298$ people watched one out of three videos on M-Turk in which a robot reports about another robot who just joined his lab
 - a) Robot is sad that there is a new robot in the lab
 - b) Robot reports without emotions
 - c) Robot is happy about the new robot in the lab
- Afterwards: Rating of the credibility and intelligence of the robot
- The sad robot is rated as less intelligent and credible

Figure 1



Sorrowful Condition

Figure 2



No Emotion Condition

Figure 3



Joyful Condition

Via <https://osf.io/r8t63>

What we “tolerate” in artificial others remains rooted in the same patterns

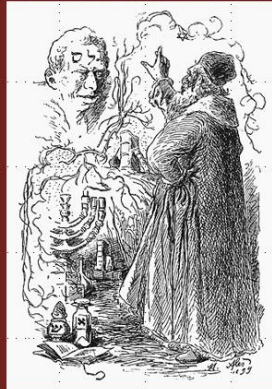
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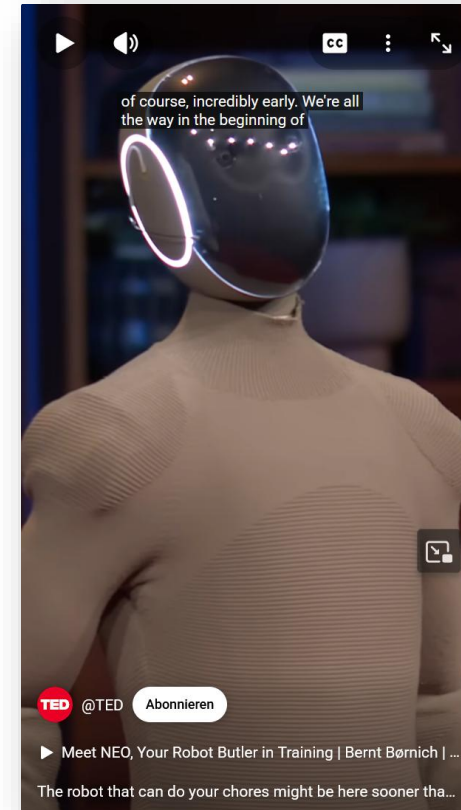
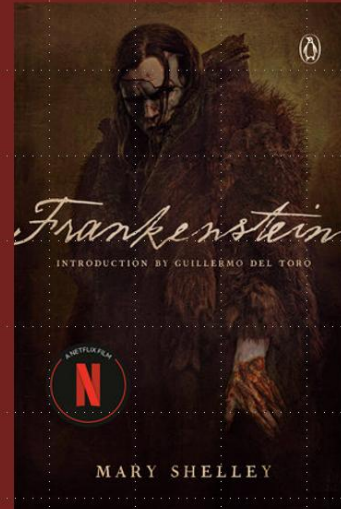
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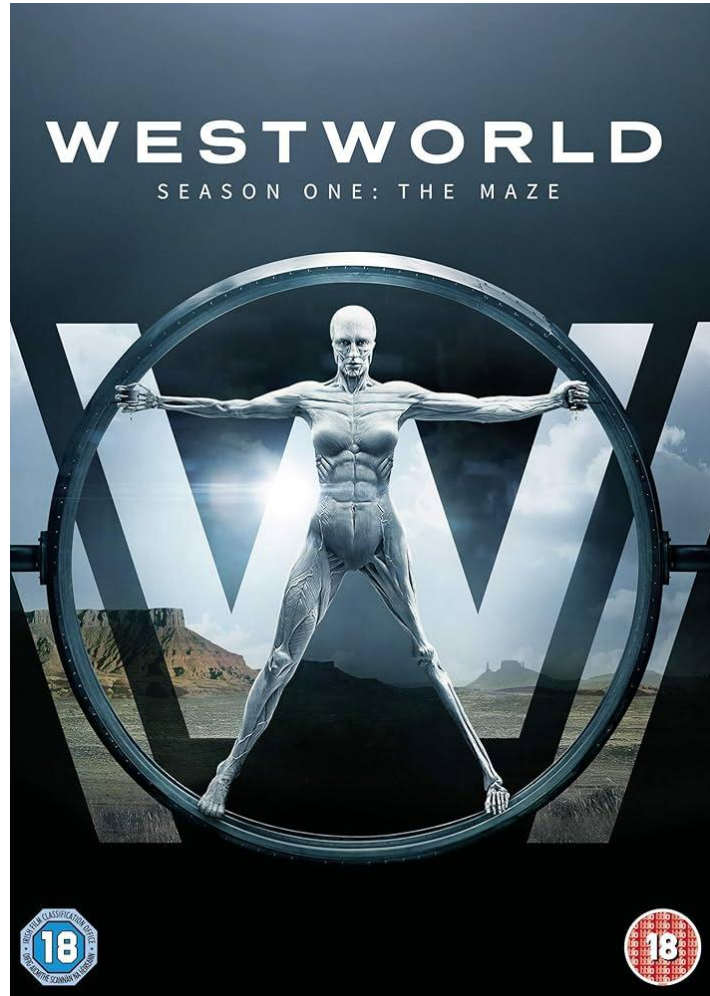
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To the horror of Frankenstein...



<https://www.youtube.com/shorts/sexqiDz8fWs>

Westworld



Westworld, Jonathan Nolan & Lisa Joy since 2016



<https://www.esquire.com/entertainment/tv/a19753403/westworld-season-1-recap/>

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**See you next week on
Thursday (or for the
supervisor meeting)**

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